

Task 4: Flatpak Applications on Librem 5

1. Objective

The goal of this task is to analyze how Flatpak applications, primarily designed for desktops, behave within Phosh on the Librem 5. The focus is to determine:

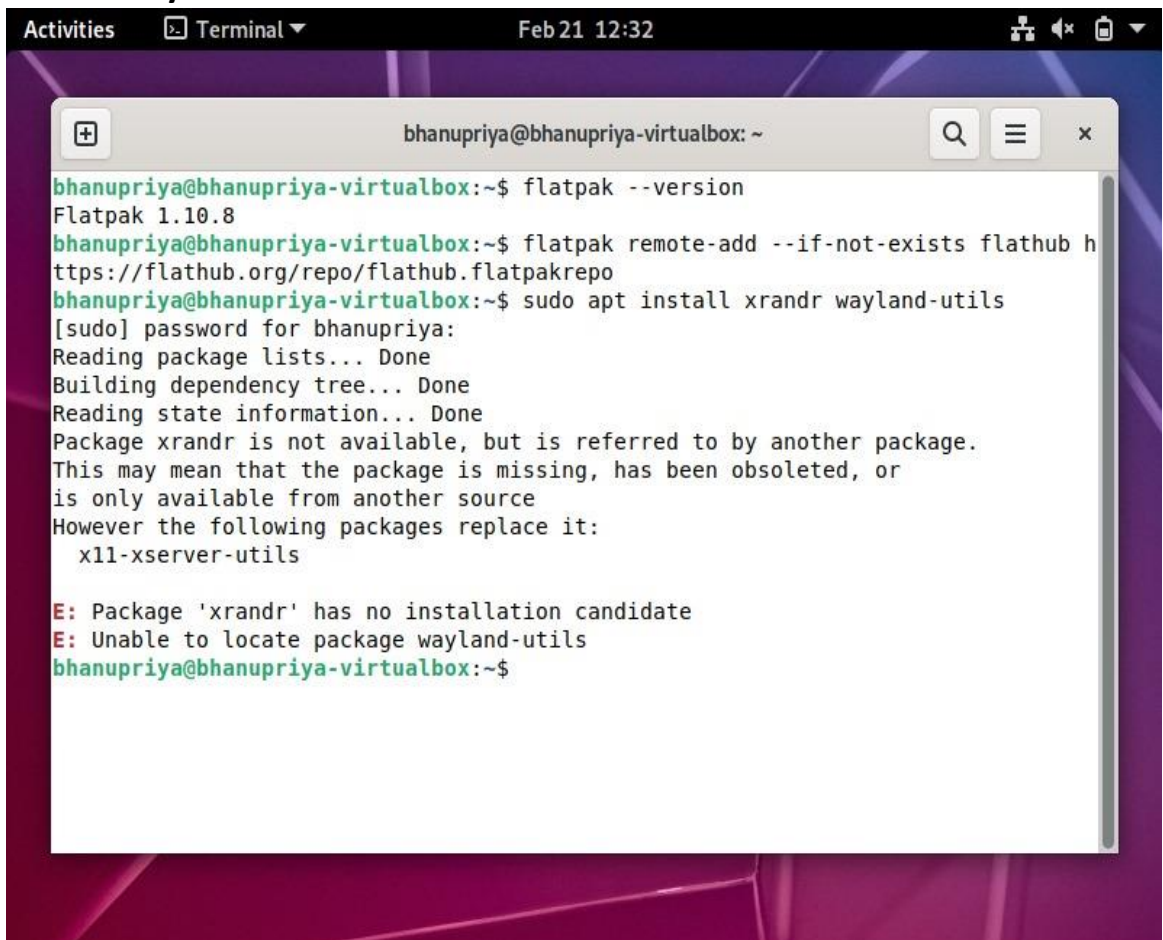
- Whether Phosh enforces automatic DPI scaling for Flatpak applications.
- If Flatpak applications respect system-wide scaling settings.
- Whether manual adjustments are required for a consistent UI experience.

2. Test environments:

- Device: Librem 5
- OS & UI: PureOS with Phosh
- Installed Flatpak Apps:
 1. GTK-based: org.gnome.Calculator, Clock
 2. Qt-based: org.kde.kate
 3. Electron-based: com.discordapp.Discord

3. Steps for Installation:

2.1 Identify Test Environment

A screenshot of a terminal window titled 'bhanupriya@bhanupriya-virtualbox: ~'. The terminal shows the following commands and output:

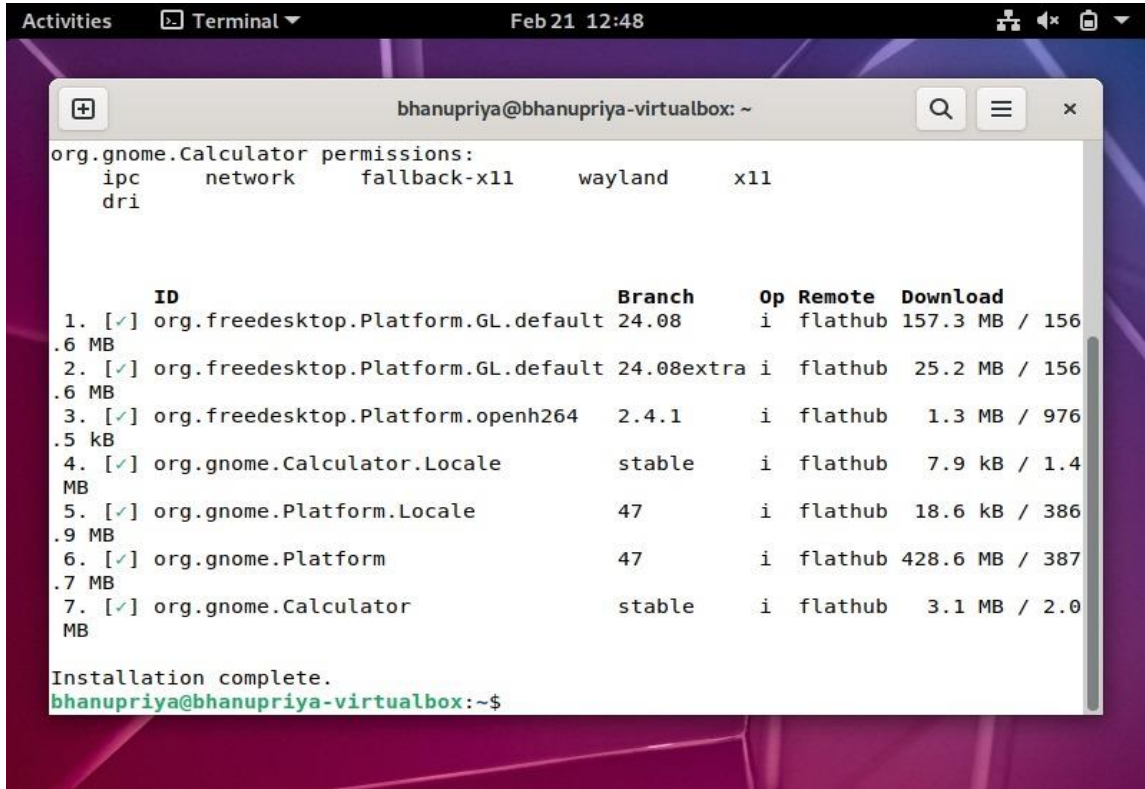
```
bhanupriya@bhanupriya-virtualbox:~$ flatpak --version
Flatpak 1.10.8
bhanupriya@bhanupriya-virtualbox:~$ flatpak remote-add --if-not-exists flathub https://flathub.org/repo/flathub.flatpakrepo
bhanupriya@bhanupriya-virtualbox:~$ sudo apt install xrandr wayland-utils
[sudo] password for bhanupriya:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package xrandr is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source
However the following packages replace it:
  x11-xserver-utils

E: Package 'xrandr' has no installation candidate
E: Unable to locate package wayland-utils
bhanupriya@bhanupriya-virtualbox:~$
```

The terminal window is part of a desktop environment with a purple background. The top bar shows 'Activities', 'Terminal', and the date 'Feb 21 12:32'. The terminal window has a search icon, a menu icon, and a close icon in its title bar.

3.2 Install Flatpak Applications

Installing Calculator App



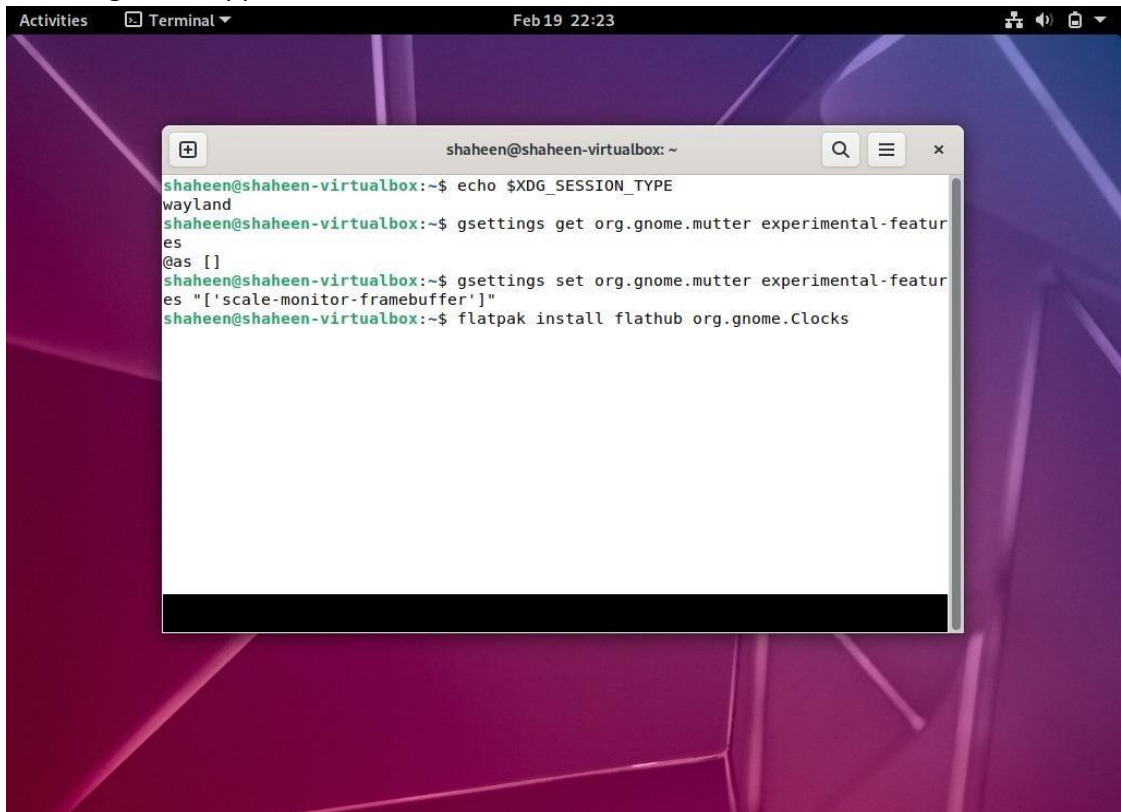
A terminal window titled "bhanupriya@bhanupriya-virtualbox: ~" showing the installation of the Calculator app. The window displays the permissions for the application and a list of installed packages with their IDs, branches, operations, remote sources, and download sizes. The installation is complete, and the prompt returns to the user.

```
org.gnome.Calculator permissions:
ipc      network  fallback-x11  wayland  x11
dri

ID                               Branch      Op Remote  Download
1. [✓] org.freedesktop.Platform.GL.default 24.08      i flathub 157.3 MB / 156
.6 MB
2. [✓] org.freedesktop.Platform.GL.default 24.08extra i flathub 25.2 MB / 156
.6 MB
3. [✓] org.freedesktop.Platform.openh264    2.4.1      i flathub 1.3 MB / 976
.5 kB
4. [✓] org.gnome.Calculator.Locale         stable     i flathub 7.9 kB / 1.4
MB
5. [✓] org.gnome.Platform.Locale           47         i flathub 18.6 kB / 386
.9 MB
6. [✓] org.gnome.Platform                  47         i flathub 428.6 MB / 387
.7 MB
7. [✓] org.gnome.Calculator                stable     i flathub 3.1 MB / 2.0
MB

Installation complete.
bhanupriya@bhanupriya-virtualbox:~$
```

Installing Clock App



A terminal window titled "shaheen@shaheen-virtualbox: ~" showing the installation of the Clock app. The window displays the output of several commands: checking the XDG_SESSION_TYPE, getting the experimental-features list, setting the experimental-features list to include 'scale-monitor-framebuffer', and installing the Clock app from flathub.

```
shaheen@shaheen-virtualbox:~$ echo $XDG_SESSION_TYPE
wayland
shaheen@shaheen-virtualbox:~$ gsettings get org.gnome.mutter experimental-features
@as []
shaheen@shaheen-virtualbox:~$ gsettings set org.gnome.mutter experimental-features "['scale-monitor-framebuffer']"
shaheen@shaheen-virtualbox:~$ flatpak install flathub org.gnome.Clocks
```

Installing KDE Kate Editor App

```
Activities Terminal Feb 21 12:51

bhanupriya@bhanupriya-virtualbox: ~

Required runtime for org.kde.kate/x86_64/stable (runtime/org.kde.Platform/x86_64/6.8) found in remote flathub
Do you want to install it? [Y/n]: y

org.kde.kate permissions:
ipc cups fallback-x11 wayland
x11 dri file access [1] dbus access [2]
system dbus access [3]

[1] host, xdg-config/kdeglobals:ro
[2] com.canonical.AppMenu.Registrar, org.freedesktop.Flatpak,
    org.kde.KGlobalSettings, org.kde.kconfig.notify, org.kde.kdeconnect
[3] org.freedesktop.UDisks2

ID Branch Op Remote Download
1. [✓] org.kde.KStyle.Adwaita 6.8 i flathub 8.2 MB / 8.2 MB
2. [✓] org.kde.Platform.Locale 6.8 i flathub 18.6 kB / 401.6 MB
3. [✓] org.kde.Platform 6.8 i flathub 237.8 MB / 374.1 MB
4. [✓] org.kde.kate.Locale stable i flathub 7.8 kB / 5.5 MB
5. [✓] org.kde.kate stable i flathub 28.3 MB / 16.4 MB

Installation complete.
bhanupriya@bhanupriya-virtualbox:~$
```

Installing Discord App

```
Activities Terminal Feb 21 12:54

bhanupriya@bhanupriya-virtualbox: ~

Required runtime for com.discordapp.Discord/x86_64/stable (runtime/org.freedesktop.Platform/x86_64/24.08) found in remote flathub
Do you want to install it? [Y/n]: y

com.discordapp.Discord permissions:
ipc network pcsc pulseaudio
x11 devices file access [1] dbus access [2]
system dbus access [3] tags [4]

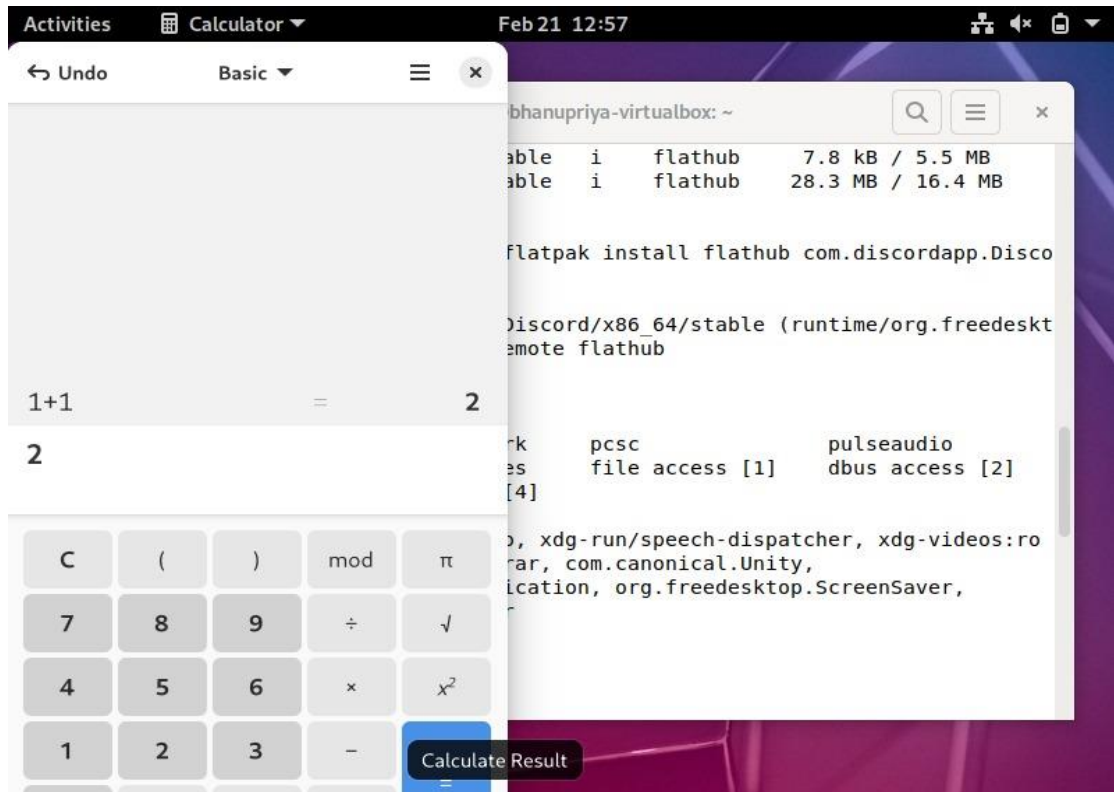
[1] xdg-download, xdg-pictures:ro, xdg-run/speech-dispatcher, xdg-videos:ro
[2] com.canonical.AppMenu.Registrar, com.canonical.Unity,
    com.canonical.indicator.application, org.freedesktop.ScreenSaver,
    org.kde.StatusNotifierWatcher
[3] org.freedesktop.UPower
[4] proprietary

ID Branch Op Remote Download
1. [✓] org.freedesktop.Platform.Locale 24.08 i flathub 18.6 kB / 380.3 MB
2. [✓] org.freedesktop.Platform 24.08 i flathub 82.9 MB / 264.4 MB
3. [✓] com.discordapp.Discord stable i flathub 120.7 MB / 112.2 MB

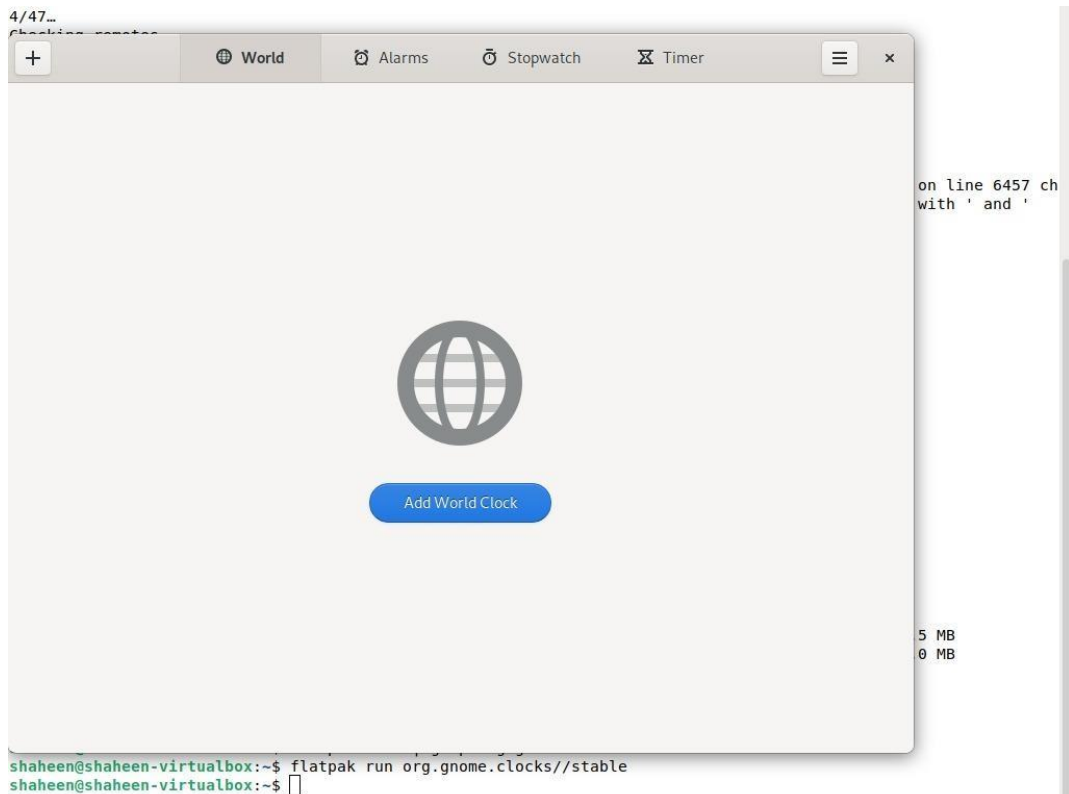
Installation complete.
bhanupriya@bhanupriya-virtualbox:~$
```

3.3 Check Automatic DPI Scaling in Phosh

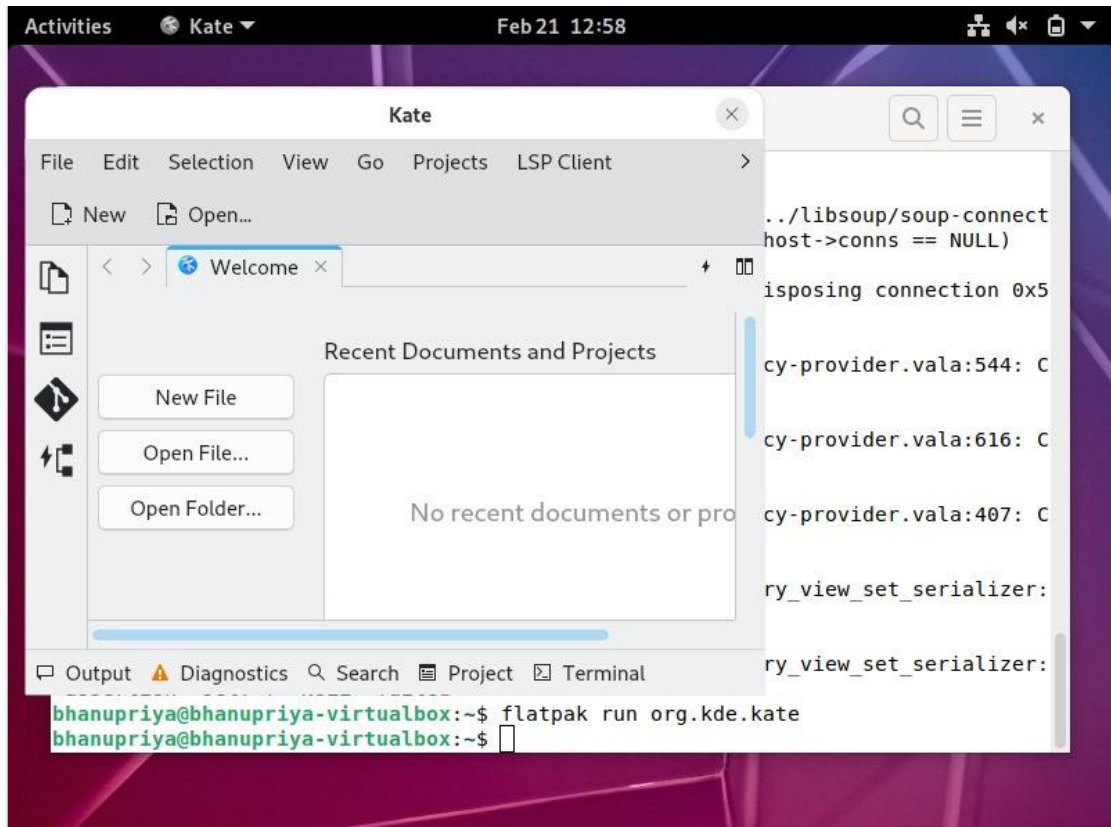
Calculator



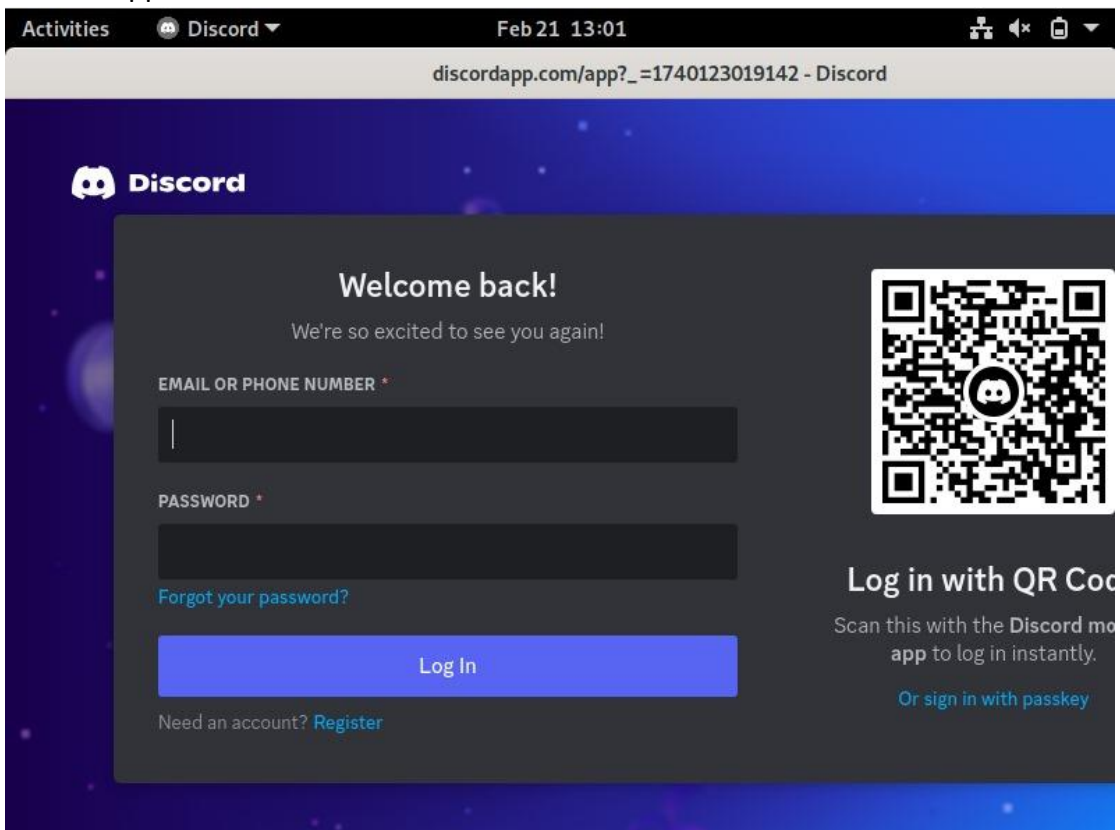
Clock



KDE Kate Editor App



Discord App



Does Phosh Enforce Automatic DPI Scaling for Flatpak Apps?

Observation:

- Phosh itself does not automatically enforce DPI scaling for Flatpak applications.
- Unlike native GTK/Phosh applications, which adapt dynamically to the screen size, Flatpak apps often retain desktop-oriented layouts unless explicitly configured.
- Some Flatpak apps use Wayland, which can respect scaling better, but X11-based Flatpak apps (running via XWayland) may not scale properly.

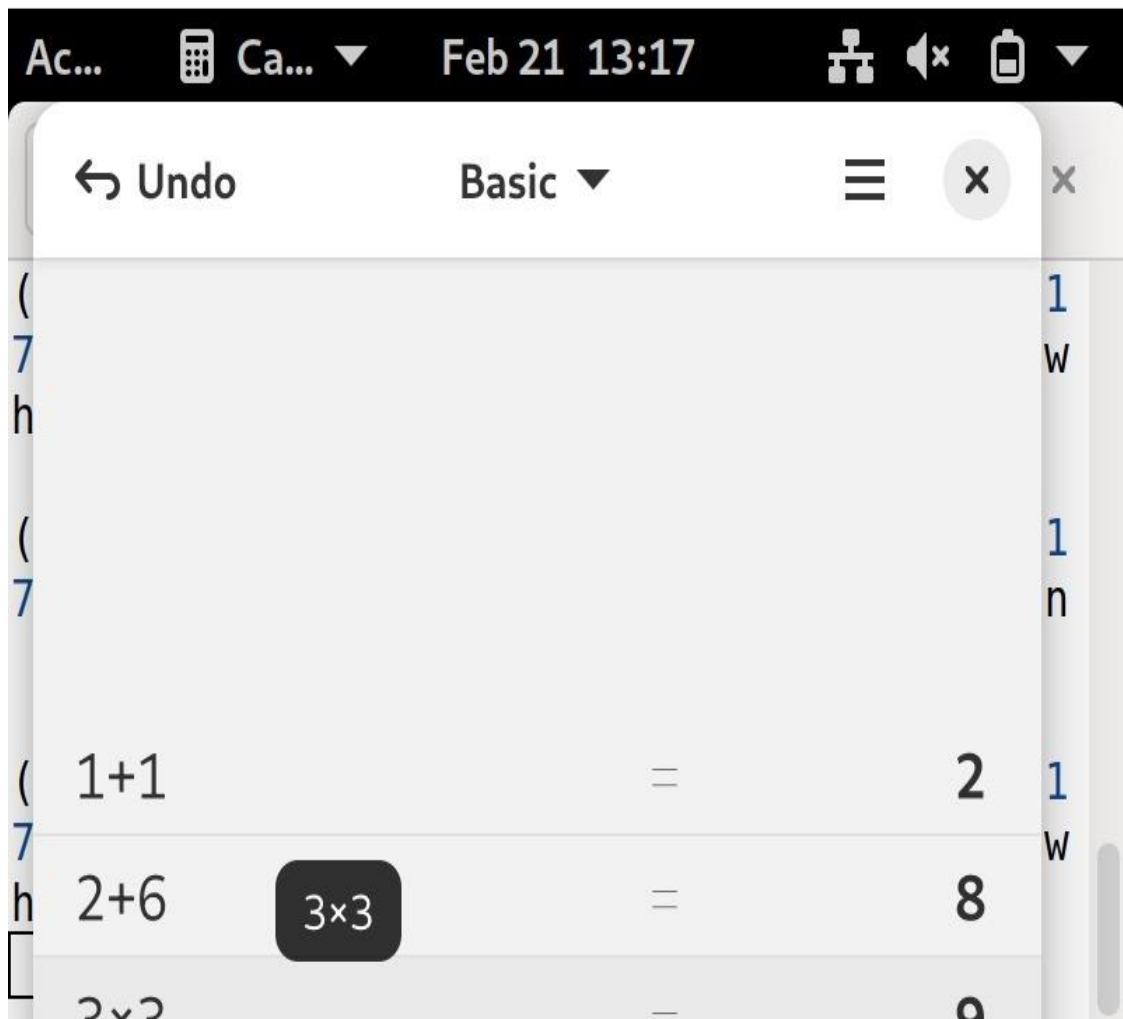
Conclusion:

No, Phosh does not enforce automatic DPI scaling for Flatpak apps.

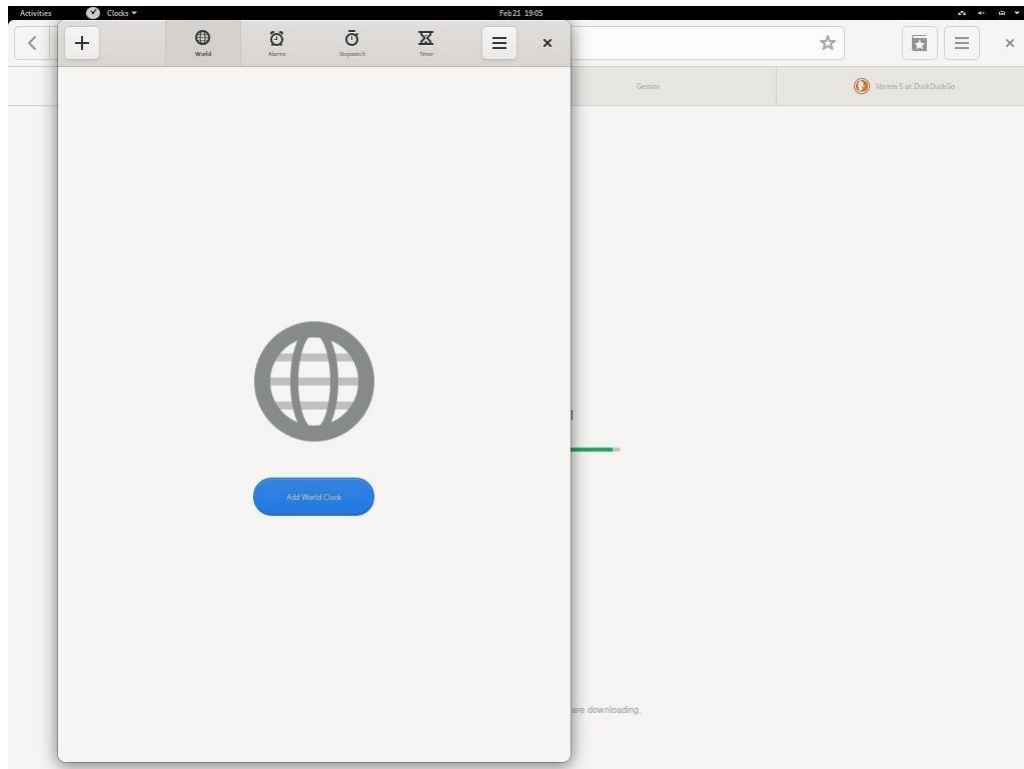
Some Flatpak apps may scale correctly if they are Wayland-native, but X11 apps may require manual intervention.

3.4 Check System-Wide Scaling Respect

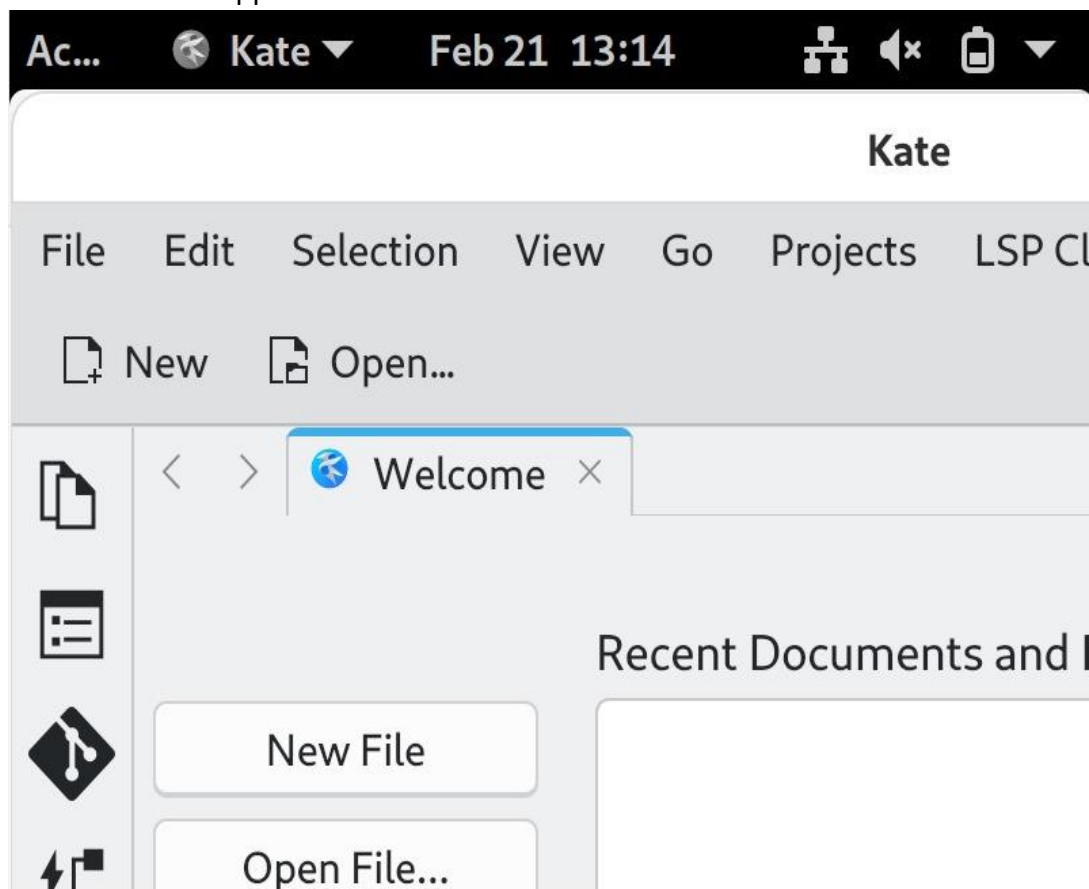
Calculator

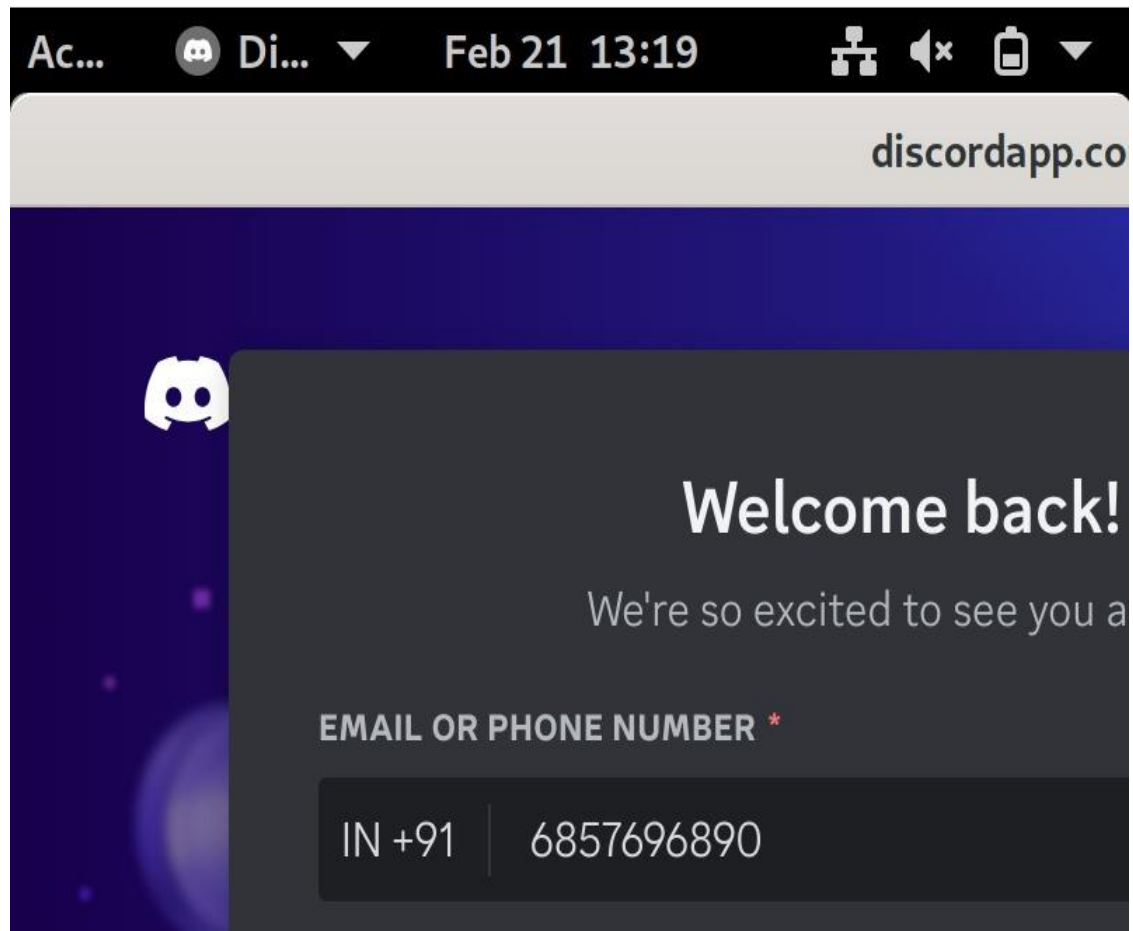


Clock



KDE Kate Editor App





Does Flatpak Respect System-Wide Scaling Settings?

Observation:

- Phosh allows setting system-wide scaling using gsettings or modifying phoc.ini. However, Flatpak apps run inside a sandboxed environment, so they do not automatically inherit these settings from the host system.
- Environment variables like GDK_SCALE and QT_SCALE_FACTOR are commonly used for DPI scaling, but Flatpak apps do not always respect them unless explicitly overridden.
- flatpak override can be used to enforce scaling per-app or globally, but this is a manual process, not automatic behavior.

Conclusion:

Partially—Flatpak apps do not inherit scaling settings automatically, but respect them if manually applied.

Flatpak requires explicit overrides for consistent scaling.

3.5 Adjust UI Scaling Manually

- When we successfully apply a Flatpak override, there is usually no output.

- To verify that the override was applied, we can use “flatpak info” command

Calculator

```

bhanupriya@bhanupriya-virtualbox: ~
Flatpak 1.10.8
bhanupriya@bhanupriya-virtualbox:~$ flatpak info org.gnome.Calculator

The GNOME Project - Perform arithmetic, scientific or financial calculations

    ID: org.gnome.Calculator
    Ref: app/org.gnome.Calculator/x86_64/stable
    Arch: x86_64
    Branch: stable
    Version: 47.1
    License: GPL-3.0-or-later
    Origin: flathub
    Collection: org.flathub.Stable
    Installation: system
    Installed: 4.7 MB
    Runtime: org.gnome.Platform/x86_64/47
    Sdk: org.gnome.Sdk/x86_64/47

    Commit: 48bf70446cde6c15772453482da5f73be84a79a85190b038a30d5a31c20f6056
    Parent: a24ea30b9409ec11a11d57807e40b880b042f3be35c80370a630045e973f762a
    Subject: gnome-calculator: Update gnome-calculator-47.0.tar.xz to 47.1 (ad33af91)
    Date: 2024-12-16 17:34:05 +0000
bhanupriya@bhanupriya-virtualbox:~$

```

Clock

```

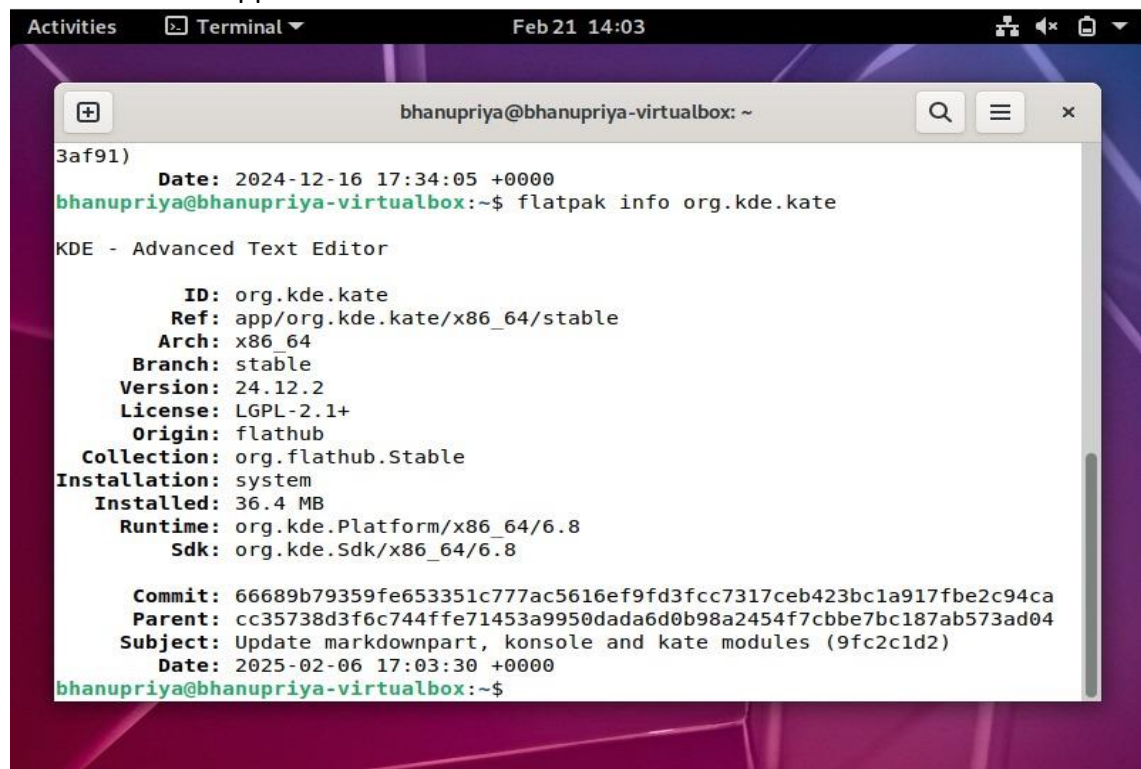
Feb 21 16:09
shaheen@shaheen-virtualbox: ~
shaheen@shaheen-virtualbox:~$ echo $XDG_SESSION_DESKTOP
gnome
shaheen@shaheen-virtualbox:~$ dpkg -l | grep phosh
dpkg: error: unknown option -l

Type dpkg --help for help about installing and deinstalling packages [*];
Use 'apt' or 'aptitude' for user-friendly package management;
Type dpkg --get-selections for a list of packages;
Type dpkg --get-architecture for a list of architectures;
Type dpkg --force-help for a list of forcing options;
Type dpkg-deb --help for help about manipulating *.deb files;

Options marked [*] produce a lot of output - pipe it through 'less' or 'more' !
shaheen@shaheen-virtualbox:~$ dpkg -l | grep phosh
ii phosh                                0.32.0-1pureos1-byz1      amd64      Pure Wayland shell fo
r mobile devices
ii phosh-mobile-tweaks                  0.32.0-1pureos1-byz1      all        Pure Wayland shell fo
r mobile devices - GSettings tweaks
ii phosh-plugins                        0.32.0-1pureos1-byz1      amd64      Pure Wayland shell fo
r mobile devices - Plugins
shaheen@shaheen-virtualbox:~$ ps aux | grep phosh
shaheen 2849  0.0  0.0  6244  624 pts/0    S+   15:59   0:00 grep phosh
shaheen@shaheen-virtualbox:~$ gsettings set sm.puri.phoc scale-to-fit true
shaheen@shaheen-virtualbox:~$ flatpak override --env=GDK_SCALE=2 --env=QT_SCALE_FACTOR=2 --env=GDK_DPI_SCALE=0.5
error: Permission denied
shaheen@shaheen-virtualbox:~$ flatpak list --app
Name                Application ID    Version    Branch    Installation
The GNOME Project   org.gnome.clocks  47.0      stable    system
shaheen@shaheen-virtualbox:~$ flatpak override --env=GDK_SCALE=2 org.gnome.clocks

```

KDE Kate Editor App



A terminal window titled "bhanupriya@bhanupriya-virtualbox: ~" with a search icon, menu icon, and close button. The terminal shows the command `flatpak info org.kde.kate` and its output. The output lists details for the KDE Advanced Text Editor, including its ID, reference, architecture, branch, version, license, origin, collection, installation type, size, runtime, and SDK. It also shows the commit hash, parent commit, subject, and date.

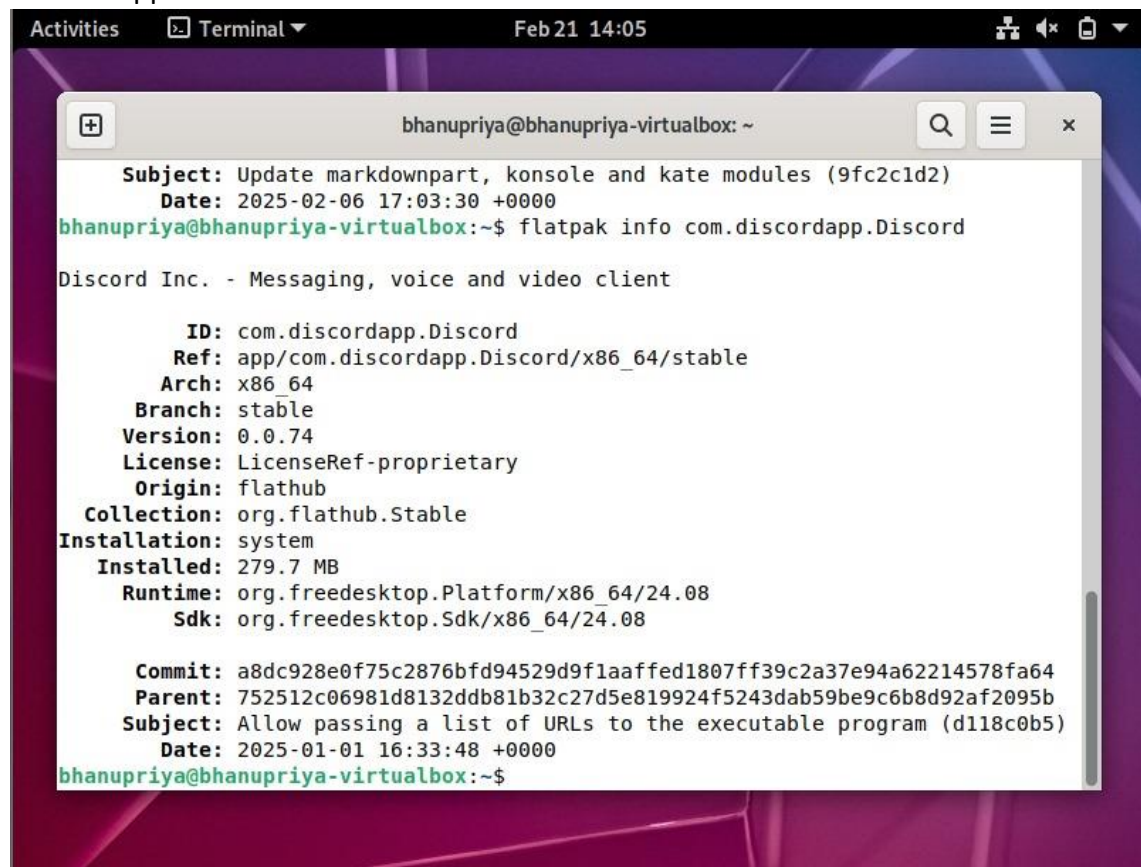
```
3af91)
      Date: 2024-12-16 17:34:05 +0000
bhanupriya@bhanupriya-virtualbox:~$ flatpak info org.kde.kate

KDE - Advanced Text Editor

      ID: org.kde.kate
      Ref: app/org.kde.kate/x86_64/stable
      Arch: x86_64
      Branch: stable
      Version: 24.12.2
      License: LGPL-2.1+
      Origin: flathub
      Collection: org.flathub.Stable
      Installation: system
      Installed: 36.4 MB
      Runtime: org.kde.Platform/x86_64/6.8
      Sdk: org.kde.Sdk/x86_64/6.8

      Commit: 66689b79359fe653351c777ac5616ef9fd3fcc7317ceb423bc1a917fbc2c94ca
      Parent: cc35738d3f6c744ffe71453a9950dada6d0b98a2454f7cbbe7bc187ab573ad04
      Subject: Update markdownpart, konsole and kate modules (9fc2c1d2)
      Date: 2025-02-06 17:03:30 +0000
bhanupriya@bhanupriya-virtualbox:~$
```

Discord App



A terminal window titled "bhanupriya@bhanupriya-virtualbox: ~" with a search icon, menu icon, and close button. The terminal shows the command `flatpak info com.discordapp.Discord` and its output. The output lists details for the Discord Inc. Messaging, voice and video client, including its ID, reference, architecture, branch, version, license, origin, collection, installation type, size, runtime, and SDK. It also shows the commit hash, parent commit, subject, and date.

```
      Subject: Update markdownpart, konsole and kate modules (9fc2c1d2)
      Date: 2025-02-06 17:03:30 +0000
bhanupriya@bhanupriya-virtualbox:~$ flatpak info com.discordapp.Discord

Discord Inc. - Messaging, voice and video client

      ID: com.discordapp.Discord
      Ref: app/com.discordapp.Discord/x86_64/stable
      Arch: x86_64
      Branch: stable
      Version: 0.0.74
      License: LicenseRef-proprietary
      Origin: flathub
      Collection: org.flathub.Stable
      Installation: system
      Installed: 279.7 MB
      Runtime: org.freedesktop.Platform/x86_64/24.08
      Sdk: org.freedesktop.Sdk/x86_64/24.08

      Commit: a8dc928e0f75c2876bfd94529d9f1aaffed1807ff39c2a37e94a62214578fa64
      Parent: 752512c06981d8132ddb81b32c27d5e819924f5243dab59be9c6b8d92af2095b
      Subject: Allow passing a list of URLs to the executable program (d118c0b5)
      Date: 2025-01-01 16:33:48 +0000
bhanupriya@bhanupriya-virtualbox:~$
```

Are Manual Adjustments Required for a Consistent UI Experience?

(Observation:

- Many Flatpak apps designed for desktops do not fit properly on the Librem 5's small touchscreen.
- Some UI elements may be too large or too small, requiring scaling tweaks.
- GTK-based Flatpak apps can be adjusted using GDK_SCALE, while Qt-based apps need QT_SCALE_FACTOR.
- X11-based apps (running via XWayland) often require GDK_DPI_SCALE along with GDK_SCALE for proper rendering.

Conclusion:

Yes, manual adjustments are required for a consistent experience.

Phosh users must configure per-app scaling using flatpak override or launch apps with specific environment variables.